Cog Sys - Question Collection

Kognitive Systeme (Technische Universität München)

Cognitive Systems: Exam questions

1 Cognition

1. Describe the Chinese room argument
   - person that does not speak chinese sits in room with input/output
   - in the room, there is a mapping of any question in chinese to any answer in chinese
   - person receives questions in chinese and uses lookup table in room to find a meaningful answer without having to understand what the question is about/having to think
   - Chinese speakers outside the room will think that the box (as seen from outside) can think as it answers meaningfully
2. Explain what is the purpose of the Chinese room argument
   - shows, that passing the Turing test does not mean that a machine can think, as person in room has no knowledge about meaning of questions and is basically a big lookup table

3. Describe the Turing test
   - person A (man) and person B (woman) are in conversation with an interrogator via text
   - person A pretends to be person B in order to deceive the interrogator
   - person A is then replaced by computer
   - if the machine can deceive the interrogator as often as the man, the computer passes the Turing test
4. Explain what is the robotics paradox:

   → Seemingly easy tasks are difficult for robots to solve (e.g., going shopping), while seemingly hard tasks that are well-defined are easy (e.g., spot welding)
5. Explain the difference between cognitivist cognitive systems and emergent cognitive systems
   - cognitivist: Cognition is used as a means of rational goal achieving, all operations the system performs are syntactic symbol operations. Cognitivist systems are based in the computational theory of mind where it is assumed that causality in semantics can be solely described through syntactic symbol operations. Learning is done through the acquisition of new knowledge
   - emergent: System is subject to continuous self-reorganization with the goal of extending its own autonomy/ to increase its space of interaction. The role of cognition in the emergent system is self-maintenance. Learning is done through the acquisition of new anticipatory skills. Embodiment is a vital part of emergent systems.
   - Shorter:
   - **Cognitivist**: based on the hypothesis that cognition is a form of computation. Cognitive functions are modeled as working computer programms.
   - **Emergent**: cognition is a continuous, self-organizing process that is driven by the interaction between the agent and its environment

6. Name seven cognitive capabilities that a system must possess in order to be self-reliant and adaptive, and interact with its environment
   - Choose 7:

- Self-Reliance: 1.Goal-Directedness, 2. Autonomy, 3. Interaction
- Adaption: 4. Reaction, 5. Learning 6. Anomaly Detection
- Perception & Action: 7. Interpretation, 8. Sensing, 9. Anticipation, 10. Action

7. Name the four different lobes in the cerebrum and explain what are their main cognitive functions
   - temporal lobe: hearing, emotion, learning, memory
   - occipital lobe: vision
   - frontal lobe: motion planning/action control/ short term memory
   - parietal lobe: body image, somatic sensation
8. Assume that you have a doll company. Your new series features dolls that are more realistic than any toy produced until now. However, the doll is highly unsuccessful and testers describe it as creepy. Name and explain the phenomenon that is taking place
   - uncanny valley: Familiarity with human-like object does not steadily increase when human-likeness increases
   - First, the familiarity rises, then, if the human-likeness grows more, the familiarity decreases again until it increases when the human likeness is again close to human-level
9. A company claims that it can make you smarter by unlocking the 90% unused capacity of your brain. Explain whether this is possible or not
   - This is a myth
   - Only 10% of brain capacity used at once as the brain has different functions that are not all needed at the same time
   - Every part of the brain is used, but they are not all used at once → this would result in epilepsy (lecture slides)
10. Define what is a cognitive system
    - autonomous system that perceives its environment, anticipates outcomes, learns from experiences, adapts to changes and is goal-driven
    - Both below are in not part of Definition on slides —> More Purpose of Cognition:
    - (system that is able to deal with uncertainty and change and operate beyond its pre-programmed behavior
    - this includes interaction with other agents, maintenance of a system state under adverse conditions via goal-achieving and learning for new action selection)
11. Define cognition
    - accumulation of system-level processing modalities, all involved in the process of knowing, consisting of different cognitive functions/processing modalities such as learning and memory
    - is not: wishing, feeling
12. Name the components of a cognitive system
    - embodiment
    - constraints
    - interaction with other agents
    - situatedness in environment
13. Name two applications that require cognition
    - Autonomous driving
    - Natural Language Processing (NLP)
    - Human-Robot Interaction
    - Physical Interaction (e.g. walking, sports)

- Collaborative Work with Multiple Agents

14. Draw the cycle of cognitive processing
    - Perception → Cognition → Action
    - Cognition: → Anticipation → Assimilation → Adaptation
15. Name the two schools of thought which deal with the mind-body problem. For each of them, describe their approach to the problem
    - monism: Mind/Body are the same substance, cognitive capabilities are directly linked to physical trait, mental states are physical states
    - dualism: Mind/Body are two different substances, this decouples mind and body and is the basis for free will
16. Explain What kinds of tasks robots/computers perform better than humans, and which ones they perform worse. Give an example for each of them.
    - Better: Well-defined tasks with environment that stay the same and is subject to few perturbations, e.g. welding
    - worse: Ill-defined tasks like shopping, where changes in the environment may change the task definition to change and there may be a lot of hard-to-predict perturbations by the environment
17. Explain why birds sing using an ultimate explanation and using a proximate explanation
    - Ultimate: Explain why: Birds want to attract mates to procreate
    - Proximate: Explain how: By singing loudly, possible mates are attracted to the bird

2 NRP and cognitive architectures

1. Name four advantages of executing experiments with virtual robots instead of physical ones.
    - Increased simulation speed
    - Change parameters to simplify experiments
    - No sensor imperfections
    - Repeat runs without deviations
    - Reset experiment automatically
    - No need for expensive robots
    - No risk of breaking robots
2. Explain what is the neurorobotics platform and explain the advantages of its usage
    - Simulation platform that allows to simulate brain functions with robots
    - It connects the domains of neuroscience, AI/ML, and robotics
    - Advantages:
        - includes lots of models
        - import own functions (from nest, TensorFlow, python, …)
        - All-in-one framework for simulation
        - HPC support
3. Name two software tools that are integrated in the neurorobotics platform
    - Gazebo: rigid body simulation for robots
    - OpenSim: modeling and simulation of movement (realistic muscle simulations, …)
    - ROS: operating system for robots (communication)
    - NEST: simulating spiking neural networks (“TensorFlow for SNNs”)
4. Explain what is the function of buffers in ACT-R
    - expose information about the current state of the module

- Every buffer stores one chunk (declarative knowledge) and is assigned to exactly one module.
- Serve as interfaces between modules.

5. Explain what is a module in ACT-R. Name two features of modules in ACT-R
   - independent parallel processing units that encapsulate specific functions.
     - Every module corresponds to one or more cognitive functions of the brain.
     - Information processing in a module is completely independent and separate from other modules
     - Every module can encapsulate arbitrarily complex sub-functions
6. Explain what is a production in R
   - Statement of a particular contingency that controls behavior

3 Human Brain

1. Explain whether the following statement is true or false: “Only 10% of the human brain is used”
   - This is a myth
   - Only 10% of brain capacity used at once as the brain has different functions that are not all needed at the same time
2. Explain what is the main purpose of the white matter in the cerebrum
   - nerve fibers connecting different cortical regions
3. Name and describe the three main anatomical features of the cerebral cortex
   - gyri: Crests that characterize the surface of the cerebral cortex
   - sulci: divide the cerebral cortex into different lobes
   - lobes: 4 lobes, each hemisphere has them
4. Name two distinctive features of human brain that makes it stand out among other species
   - lower membrane capacity of neurons: Better electric signal transmission
   - Neuron morphology: More spine-rich sophisticated dendritic trees
   - Sequence processing: superior capabilities of learning sequences of stimuli
   - Highest encephalization quotient
   - highest neuronal density in cerebral cortex
5. Explain what is the Blue Brain Project
   - data-driven reconstruction process for cortical microcircuits, modeling neurons and synapses
   - (alternative: biggest attempt to simulate brain a brain by extracting 14.000+ neurons from rats and simulating them with supercomputers)
6. Provide the names of the three main components of a biological neuron and describe their main tasks.
   - soma: Neuron body containing the nucleus, creating action potentials from stimulus
   - dendrites: receive signal from pre-synaptic neuron
   - axons: signal transmission from soma towards post-synaptic neurons
7. The cerebral cortex is structured into four different lobes. Name three different lobes present in the cortex. For each of them, name one cognitive function that it implements.
   - temporal lobe: hearing, emotion, learning, memory
   - occipital lobe: vision
   - frontal lobe: motion planning / action control / short term memory
   - parietal lobe: body image, somatic sensation
8. Name and describe the two subsystems of the vertebrate nervous system
   - CNS: brain + spinal cord, collecting and distributing data throughout the body

- PNS: Connect organs, muscles to CNS

9. Explain the function of the hippocampus
   - formation of episodic long-term memories
10. Name two features or properties of the peripheral (autonomic) nervous system
    - sympathetic nervous system: connecting organs to brain, prepare body for stress (e.g., increase heart rate and blood flow, sweat due to heat)
    - parasympathetic nervous system: set body to resting state, increase digestive functions
11. Map the nervous system into the perception-cognition-action diagram
    - action: spinal cord and muscles
    - cognition: brain
    - perception: sensory organs
12. Explain what is the main purpose of the brainstem. Name the four section that compose it
    - Purpose: connect cerebrum with spinal cord
    - 4 sections:
      - Diencephalon
      - Midbrain
      - Pons
      - Medulla Oblongata
13. Describe what is the encephalization quotient
    - difference of the brain size of an animal from the base-prediction – humans have the largest
14. Briefly explain the function of the amygdala in the human brain
    - analyze motivational/emotional significance of stimuli

4 Biological Neural Networks

1. Name three learning rules based on Hebbian learning
   - Covariance rule
   - Correlation rule
   - BCM theory
   - Oja's Rule
   - STDP
2. Name and explain two paradigms for encoding information in spiking neural networks
   - rate coding: The information is carried in the averaged spike rate within a certain time window (e.g. muscle excitation)
   - Time-to-first spike coding: The information is encoded in the time of emission of the first spike after a reference point.
   - Coding by Synchrony or Correlation (from E04 p51+52)
3. Name five levels of modeling the dynamics of biological neurons
   - detailed compartmental model
   - reduced compartmental models
   - single-compartment model
   - cascade models
   - black-box models
4. Explain what is the action potential of a biological neuron
   - spike potential that neuron emits in case it fires
   - it is not dependent on the charges flowing into neuron before → always has the same shape (all or none event)

- potential rises during depolarization and falls during hyperpolarization

5. Describe the Hodgkin-Huxley Model. You may draw a diagram
    - Neuron model
    - Model membrane capacity via capacitor and ion currents (leakage current, potassium and sodium)
    - model Nernst potential via batteries
    - complex non-linear resistors model the dynamics of opening and closing the ion channels
6. Explain the meaning of the statement “Neurons that fire together, wire together” in the context of learning. Name which type of learning is based on this statement
    - core concept of hebbian learning
    - change of synaptic weights depends on pre/post synaptic activity
    - Therefore, synapses between neurons that have a high correlation in the firing pattern and have high pre/post synaptic activities at the same time get reinforced
    - If interconnected neurons become active very close in time during a particular event, their connection strengthens and a memory of this event is “formed”
7. Explain what is neuroplasticity and why it is relevant for the function of the central nervous system
    - adaptive change in structure and function of nervous system
    - relevant as it enables learning
8. Explain what is the refractory period of a neuron. Name the two main phases that it can be split into
    - Period after hyperpolarization during which the neuron regenerates its action potential
    - subdivision:
        - absolute: Firing again impossible as the sodium channels are blocked
        - relative: Firing again possible, but stronger activation needed
9. Explain what is the resting membrane potential of a neuron
    - voltage between inside and outside of neuron at equilibrium between chemical gradient and voltage gradient → it is the voltage gradient at this point
10. Name four different activation functions for general analog neuron models
    - Radial basis function
    - ReLU
    - Sigmoid
    - Tanh
11. Explain what are the main differences between analog neuron models and spiking neuron models
    - Analog: continuous output value, easy to compute via weights and inputs to the neuron as well as activation function
    - Spiking: More close to behavior of physical neuron, emits digital spike trains instead of continuous output value
12. Draw a plot of the membrane potential of the neuron over time and label the different sections.
    - Resting period
    - depolarization (peaks with action potential)
    - hyperpolarization
    - refractory period

13. Explain the function of the myelin sheath around nerve fibers
    - increase membrane resistance in order to increase conductivity of axon
14. Solve the differential equation of a LIF neuron for a given scenario.
    - Left as an exercise to the reader – see exercise 5 problem 1
15. Name two different types of neural plasticity and briefly describe them.
    - functional plasticity: change functions of neurons
    - structural plasticity: create/delete synapses

5 Machine Learning

1. Name three methods for avoiding the problem of overfitting
    - data augmentation
    - L2 regularization
    - dropout

From lecture:

- Regularization
- More Training Data
- Dataset Augmentation

2. In the context of machine learning, describe the concept of Max-Pooling.
    - Feature selection by sliding a kernel over the input feature map that maps all inputs to one output, which is the maximum of all kernel inputs
3. Define “machine learning” with your own words. Provide an example of a robotic task that does not require machine learning.
    - ML: Learning to approximate some underlying process in the data at hand to, e.g., cluster it or retrieve I/O relationships, enables system to exhibit behavior beyond pre-programmed capabilities
4. Explain what is the advantage of converting the convolutional kernels and inputs of a neural network into the Fourier Space. Explain What is the main challenge for this approach
    - Advantage: Convolutions become simple multiplications in Fourier space. The number of operations required for computing convolutions can be reduced.
    - Challenge: keep the computational cost for the Fourier transformation low
5. Explain the difference between interpolation and regression.
    - Interpolation: function goes through all points
    - regression: try to learn underlying function and predict outputs in accordance to this, generalized better with new samples
6. State the problem of using least squares for linear classification. Explain how support vector machines address this problem
    - Linear classification: Prone to be affected by outliers, as the separating hyperplane is greatly influenced by those
    - SVM: Maximize margin between classes, making the classification more robust to outliers
7. State the formula of the output of a general analog neuron model and calculate the output given the parameters.
    - y = A(w_1*x_1 + w_2*x_2 + … w_n*x_n + b)
8. Name three learning paradigms. For each of them, describe the type of data needed for training.
    - Supervised (Data: pairs of x, y) labeled data
    - unsupervised: Only x unlabeled data

- semi-supervised: Part of the data labeled, part not → need to have underlying assumptions of the data (such as: they all belong to the input distribution)

9. Explain the method of k-fold cross-validation for partitioning a dataset.
   - Split train data into k folds
   - train k times, each time validating on different fold
   - compute validation error on different folds by averaging over the respective validation errors

10. Describe the principle of Occam's Razor and state how it is applied in machine learning.
   - Always choose the simplest hypothesis that explains some process
   - In ML: Always choose the model with the fewest number of parameters

11. Name and describe the role of the two streams of visual processing
   - dorsal: Explain "Where" for objects and "how" for motions
   - ventral: Explain "what" about objects

12. Name and describe the three conceptual processing layers in the human visual system.
   - Lines
   - orientations
   - objects
   - Lower-Level Vision: basic image analysis (e.g. lines)
   - Mid-Level Vision: processing of objects, surfaces, and orientations
   - High-Level Vision: visual processing based on memory and intentions

13. Name a problem that can not be solved by a single perceptron with a linear activation function. State briefly or draw the reason for that.
   - XOR problem
   - Not linearly separable, perceptron is only able to classify linearly separable classes

14. State the difference between regression and classification
   - regression: predict continuous output value
   - classification: Classify input into different distinct classes

6 Reinforcement Learning

1. Explain the difference between on-policy and off-policy learning
   - on-policy: target policy is behavior policy, sample-based version of solving the bellman equation for the action value functionw
   - off-policy: target policy is not behavior policy, solve the exploration problem by acting greedily to obtain the state action value for the next state and only sampling current state action value, solves the bellman optimality equation using sample based approach

2. Briefly explain what "bootstrapping value" means in the context of reinforcement learning.
   - Knowing the MDP, one can bootstrap the value function by iteratively estimating the value function based the previous step's value function → estimate optimal value function
   - No interaction with the environment is needed during bootstrapping

3. Name and describe the two alternating steps that are performed iteratively under the framework of Generalized Policy Iteration.
   - Iterative policy evaluation and improvement to find optimal value function and policy
   - policy evaluation → Find value function corresponding to current policy
   - policy improvement → Select policy based on value function

4. Describe and state the differences between classic conditioning and operant conditioning in the context of associative learning processes
   - Classic conditioning:13. Name two applications that require cognition

        - Unconditioned Stimulus: Stimulus that produces a reaction in a subject
        - Conditioned reaction: This reaction is triggered by unconditioned stimulus, through classical conditioning, i.e., presenting the unconditioned stimulus always together with the conditioned stimulus, the conditioned stimulus will at some point lead to the conditioned reaction without the unconditioned stimulus
    - Operant conditioning: Use stimulus (positive/negative) as a response to a certain behavior → This either strengthens/weakens the exhibitance of the behavior

5. State the reason behind using a linear activation function in the output layer of a Deep Q-Network.
    - Not a classification but a regression task as Q-function estimation is a regression problem → linear functions are suited for regression problems

Guest Lecture

1. Name the two main types of modern neuroimaging. Name one example for each of them
    - structural neuroimaging: Diffusion tensor imaging (DTI)
    - functional neuroimaging: Electroencephalography (EEG)

2. Explain the difference between MRI and fMRI, and how they help to better understand neural activity
    - MRI: structural neuroimaging, detects different cytoarchitectonics due to the different magnetic dissipation properties of hydrogen molecules in different surrounding tissue
    - fMRI: functional neuroimaging with high spatial resolution. Measures oxygen concentration in the brain that changes due to neural activity